

- 2 A water jetpack consists of a backpack consisting of a motor, two thrusters and a pipe for constant supply of water. A diagram of one thruster is shown in Fig. 2.1.

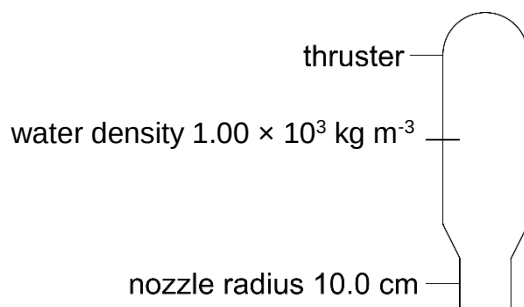


Fig. 2.1

Water is ejected vertically downwards through the nozzle. The thruster moves vertically upwards. The density of water is $1.00 \times 10^3 \text{ kg m}^{-3}$. The nozzle has a circular cross-section of radius 10.0 cm. The thruster ejects 30.0 kg of water in 0.20 s. Assume that the water leaving the nozzle has the shape of a cylinder of radius 10.0 cm.

- (a) (i) Show that the water has a constant speed of 4.8 m s^{-1} relative to the thruster. [2]

$$\rho = \frac{m}{V}$$

$$m = \rho \pi r^2 L$$

$$\frac{m}{t} = \rho \pi r^2 v$$

$$\frac{30.0}{0.20} = 1000 \times \pi \times 0.100^2 \times v \quad [1]$$

$$v = 4.8 \text{ m s}^{-1}$$

- (ii) Show that the force exerted on water ejected by the jetpack is $1.4 \times 10^3 \text{ N}$. [1]

$$F = \frac{\Delta p}{\Delta t}$$

$$= \frac{30 \times 4.8}{0.20} \quad [1] = 700 \text{ N}$$

$$\text{Force on water by jet pack} = 2 \times 700 = 1400 \text{ N}$$

- (iii) Determine the maximum combined mass of a jetpack with its user that can be lifted with an acceleration of 9.81 m s^{-2} . Explain your answer.

By Newton's 3rd law of motion, the force on jet pack with user by water is equal in magnitude and opposite in direction to the force on water by the jet pack with user. [1]

Applying Newton's 2nd law of motion on the jet pack with user,

$$\sum F_{\text{jet+user}} = m_{\text{jet+user}} a_{\text{jet+user}}$$

$$F_{\text{on jet+user}} - m_{\text{jet+user}} g = m_{\text{jet+user}} a_{\text{jet+user}}$$

$$m_{\text{jet+user}} (a_{\text{jet+user}} + g) = F_{\text{on jet+user}}$$

$$m_{\text{jet+user}} (9.81 + 9.81) = 1400 \quad [1] \rightarrow m = 71 \text{ kg} \quad [1]$$

mass = kg [3]

- (b) When in use, the jetpack takes in a constant volume of water from the environment and has a constant mass. To move horizontally, the user tilts the thrusters slightly using the handles connected. Fig. 2.2 shows one such jetpack in use above a lake.

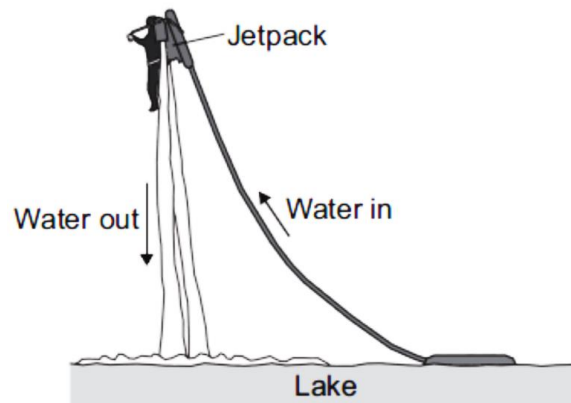


Fig. 2.2

- (i) Determine, for a user with a jetpack of combine mass 80 kg, the maximum angle with respect to the vertical which the thrusters can be tilted without the user losing altitude. Assume that the force exerted by the water pipe on the user is negligible.

$$\sum F_y = 0$$

$$F \cos \theta - mg = 0$$

$$\cos \theta = \frac{80 \times 9.81}{1400} \quad [1]$$

$$\theta = 56^\circ \quad [1]$$

maximum angle = ° [2]

(ii) Calculate the power delivered to the water for a stationary user just above the water.

$$\begin{aligned}
 P &= \frac{\text{gain in KE of water}}{t} \\
 &= \frac{\frac{1}{2}(30.0)(4.8)^2}{0.20} \times 2 \quad [1] \\
 &= 3400 \text{ W} \quad [1]
 \end{aligned}$$

power delivered = W [2]

[Total: 10]

3 (a) Two inclined planes PA and LA each have the same constant gradient. They meet at their